

Exterior balls are subcritical for every distance weight

A full negative answer to Open Problem (3) of arXiv:2303.03527

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Status. Candidate full counterexample, likely valid; human review recommended. The proof settles the source’s endpoint question negatively and, on exterior balls, proves the stronger result for every real weight exponent. Novelty confidence is moderate because the source archive contains a discarded planar seed after the final document; this is disclosed precisely below.

1 Source question

The source is U. Das, Y. Pinchover, and B. Devyver, *On existence of minimizers for weighted L^p -Hardy inequalities on $C^{1,\gamma}$ -domains with compact boundary*, arXiv:2303.03527, published in *Journal of Spectral Theory* 15 (2025), 1089–1138[1]. Open Problem (3), on printed page 32, asks:

Let Ω be a $C^{1,\gamma}$ -exterior domain and $\alpha + p \in \{1, N\}$. Is $-\Delta_{\alpha,p}$ critical in Ω ?

Green function of the p -Laplacian does not satisfy the Hopf-type lemma on C^∞ -domains, see [25, Section 3.2]. One may also observe that the $C^{1,\gamma}$ -regularity of the domain is essential for the estimates of the weighted Hardy constant at infinity which plays a significant role in the gap phenomenon.

Finally we propose three open problems:

- (1) Does the gap phenomenon hold in domains with compact C^1 -boundary?
- (2) For $1 < \alpha + p < N$ with $N < 7$, are there $C^{1,\gamma}$ -exterior domains with a positive spectral gap? (cf. Remark 7.7).
- (3) Let Ω be $C^{1,\gamma}$ -exterior domain and $\alpha + p \in \{1, N\}$. Is $-\Delta_{\alpha,p}$ critical in Ω ?

APPENDIX A. A CHAIN RULE

In this appendix we prove a chain rule for $\mathcal{I}_p$. In the case $p \geq 2$, the mapping $t \mapsto |t|^{p-2}t$ is of class C^1 , so the chain rule for $\mathcal{I}_p$ simply follows from the standard chain rule in $W^{1,p}$. In the case $p < 2$, one needs to take care of the singularity at zero of $t \mapsto |t|^{p-2}t$. We prove the chain rule in the framework of a smooth manifold M . For our purposes one should take

Figure 1: Real full-width crop of printed page 32 of the source paper.

2 Definitions and proof intuition

For $N \geq 2$, $1 < p < \infty$, and a domain Ω , let $\delta_\Omega(x) = \text{dist}(x, \partial\Omega)$ and

$$-\Delta_{\alpha,p}u := -\text{div}\left(\delta_\Omega^{-\alpha}|\nabla u|^{p-2}\nabla u\right).$$

The source calls the associated nonnegative functional *critical* when no nonzero compactly supported nonnegative potential can be subtracted while preserving nonnegativity; otherwise it is *subcritical*. It recalls the equivalent criterion that a nonnegative operator is critical exactly when its equation has, up to multiplication by a positive constant, a unique positive supersolution (then a solution) [1, Section 2].

Proof intuition. An exterior ball is radial, and a radial weighted p -harmonic function is governed by conservation of flux. Prescribing unit flux determines its derivative explicitly. One can integrate that derivative from whichever end is integrable: from the boundary when $\alpha > 1 - p$, and from infinity when $\alpha \leq 1 - p$. This always yields a positive nonconstant solution. Since the constant function is another positive solution, the uniqueness required for criticality fails. The endpoint cases in the question are merely the two borderline logarithmic instances of this all-weight construction.

3 The all-weight counterexample

Theorem 1. *Let $N \geq 2$, $1 < p < \infty$, $R > 0$, and*

$$\Omega_R := \mathbb{R}^N \setminus \overline{B_R(0)}, \quad \delta_{\Omega_R}(x) = |x| - R.$$

For every $\alpha \in \mathbb{R}$, the operator $-\Delta_{\alpha,p}$ is subcritical in Ω_R . In particular, for every such N, p, R it is subcritical at both $\alpha + p = 1$ and $\alpha + p = N$.

Proof. Write $r = |x|$ and define, for $r > R$,

$$b_\alpha(r) := (r - R)^{\alpha/(p-1)} r^{-(N-1)/(p-1)}.$$

We construct a finite positive radial function by choosing the integrable end:

$$U_\alpha(r) := \begin{cases} \int_R^r b_\alpha(s) ds, & \alpha > 1 - p, \\ \int_r^\infty b_\alpha(s) ds, & \alpha \leq 1 - p. \end{cases} \quad (1)$$

In the first branch, the integrand is asymptotic to a positive constant times $(s - R)^{\alpha/(p-1)}$ near R , and the exponent is greater than -1 ; therefore the integral is finite. In the second branch,

$$b_\alpha(s) \asymp s^{(\alpha - N + 1)/(p-1)} \quad (s \rightarrow \infty).$$

Here $\alpha \leq 1 - p < N - p$, so the exponent is strictly less than -1 and the tail integral is finite. Thus U_α is finite and strictly positive on (R, ∞) . It is smooth locally in that interval and

$$U'_\alpha(r) = \sigma b_\alpha(r), \quad \sigma := \begin{cases} 1, & \alpha > 1 - p, \\ -1, & \alpha \leq 1 - p. \end{cases}$$

For a radial function, the weighted radial flux is

$$(r - R)^{-\alpha} r^{N-1} |U'_\alpha(r)|^{p-2} U'_\alpha(r).$$

Since $b_\alpha(r)^{p-1} = (r - R)^\alpha r^{-(N-1)}$, this flux is the constant σ . Consequently,

$$\operatorname{div}((r - R)^{-\alpha} |\nabla U_\alpha|^{p-2} \nabla U_\alpha) = \frac{1}{r^{N-1}} \frac{d}{dr}(\sigma) = 0 \quad \text{in } \Omega_R. \quad (2)$$

The function U_α belongs to $W_{\text{loc}}^{1,p}(\Omega_R)$, is positive, and is nonconstant because its radial derivative never vanishes.

The constant function 1 is also a positive solution. In particular, the Agmon–Allegretto–Piepenbrink criterion quoted in the source makes the functional nonnegative. If $-\Delta_{\alpha,p}$ were critical, the source’s criticality criterion would force all positive solutions (indeed all positive supersolutions) to be proportional. The functions 1 and U_α are not proportional, a contradiction. Hence the operator is subcritical. $\square$

The two requested endpoints

At $\alpha + p = N$, formula (1) uses the boundary end and gives

$$U_N(r) = \int_R^r (s - R)^{(N-p)/(p-1)} s^{-(N-1)/(p-1)} ds;$$

its integrand is asymptotic to s^{-1} at infinity, so U_N has logarithmic growth. At $\alpha + p = 1$, formula (1) uses the infinite end and gives

$$U_1(r) = \int_r^\infty (s - R)^{-1} s^{-(N-1)/(p-1)} ds;$$

it has logarithmic blow-up only as $r \downarrow R$. Both are nevertheless positive local weak solutions in the open exterior domain, exactly as required by the source definitions.

4 Verification, scope, and novelty

The accompanying script checks the constant-flux identity and both convergence branches at 675 representative parameter/radius tuples, including $p < 2$ and both endpoint families. It passes. This is a regression check, not part of the proof.

The result is a full negative answer if the source question asks whether criticality holds in general for all $C^{1,\gamma}$ exterior domains: every exterior ball is a counterexample. It does not classify which nonradial exterior domains might be critical if the question was intended as a domain-by-domain classification.

A bounded search of the four run indexes and exact/near-exact web and arXiv phrases found no later paper explicitly answering Open Problem (3), nor the all- α exterior-ball theorem. However, the downloaded arXiv source is unusual: after the current document’s first `\end{document}`, a discarded older document records the special case $(N, p, \alpha) = (2, 2, 0)$ with solutions 1 and $\log r$, and says a similar argument should work generally. That passage is absent from the final PDF and gives neither the two full endpoint families nor the all-weight proof above, but it establishes prior author awareness of the planar seed. Mathematical confidence is therefore high and novelty confidence moderate. Human review should chiefly assess this provenance caveat and the intended quantifier in the source question.

References

- [1] U. Das, Y. Pinchover, and B. Devyver, *On existence of minimizers for weighted L^p -Hardy inequalities on $C^{1,\gamma}$ -domains with compact boundary*, J. Spectr. Theory **15** (2025), 1089–1138; arXiv:2303.03527.